

# David Dyka

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## EDUCATION

### Rensselaer Polytechnic Institute

B.S. Aerospace Engineering (Candidate)

Troy, NY

Expected May 2027

**Relevant Coursework:** Aerodynamics; Aerospace Structures & Materials; Propulsion Systems; Spaceflight Mechanics; Modeling & Control of Dynamic Systems; Fluid Mechanics; Numerical Methods & Programming; Manufacturing Processes & Systems

## SKILLS

**CAD & Analysis:** SolidWorks, Siemens NX, Fusion 360, GD&T, Engineering Drawings, Structural Analysis, Finite Element Analysis

**Manufacturing & Design:** Design for Manufacturability, Additive Manufacturing, CNC Machining, Shop Fabrication, Tooling & Fixture Design, 5S, Process Improvement, Time Studies

**Programming:** MATLAB, Python, Java, Arduino, Experimental Data Collection & Analysis

**Experimental Systems:** Sensor Integration, Automated Test Systems, Piezoelectric Sensing, Design Verification, Technical Documentation

## PROFESSIONAL EXPERIENCE

### Atlas Copco

Manufacturing Engineering Intern

Troy, NY

June 2026 – Present

- Supporting a major facility expansion by modeling shopfloor and quality-control areas in CAD, enabling engineers and production staff to evaluate equipment placement, workflow, and floor-space utilization before physical layout changes.
- Created CAD models and engineering drawings for three large production skid plates after a shop-floor safety incident, supporting root-cause documentation and future design improvements.
- Designed, fabricated, and validated a custom ratchet fixture to replace a failed shop-floor tool holder, using CAD, additive manufacturing, and finite element analysis to deliver a durable production-ready solution.
- Conducted 5S audits and shop-floor process observations to identify opportunities for improved safety, organization, ergonomics, and manufacturing efficiency.
- Supported capital equipment justification for an additional 3D printer by contributing technical content and presentation materials for an investment team review.

### Rensselaer Polytechnic Institute — Undergraduate Research

Undergraduate Researcher, Aerosol Jet Printing

Troy, NY

January 2026 – May 2026

- Produced stochastic fluorescent deposition patterns using aerosol jet printing to support physically unclonable, counterfeit-resistant material authentication for defense and aerospace supply-chain traceability.
- Improved experimental repeatability and data quality by designing precision fixturing and an Arduino-controlled automated masking system, eliminating positional and timing variability across test runs

### Polymathic Innovations

Manufacturing Engineering Intern

Troy, NY

January 2026 – May 2026

- Led scale-up of a modular micro wind turbine line from pilot to small-batch production, cutting unit cost from \$5,000 to \$3,000 through DFM redesign, material reselection, and local vendor coordination

Mechanical Engineering Intern

September 2025 – December 2025

- Delivered structurally validated, code-compliant rooftop and ground-mount systems for residential micro wind turbines by performing load analysis against operational and environmental conditions, incorporating industry standards and installer feedback under a compressed timeline

## PROJECTS

### Structural Health Monitoring — Piezoelectric Sensing System

- Demonstrated non-destructive defect detection in structural materials by instrumenting specimens with piezoelectric crystals, transmitting ultrasonic pulses, and analyzing signal attenuation patterns to identify internal damage — directly mirroring SHM techniques used in aerospace airframe inspection and fatigue monitoring

### CubeSat Drag Sail Deorbit Sizing Tool

- Developed a MATLAB orbital decay model and drag sail sizing calculator to evaluate CubeSat deorbit strategies across varying spacecraft mass, altitude, and sail parameters — producing a reusable design tool applicable to low-Earth orbit debris mitigation planning

### Formula 1 Front Wing — Finite Element Analysis

- Evaluated stress distribution, deformation, and structural margins of a Formula 1 front wing under aerodynamic loading by performing FEA in Siemens NX, applying aerospace-standard analysis methods to a high-performance structural design problem

## ADDITIONAL INFORMATION

**Clearance Eligibility:** U.S. Citizen; eligible for U.S. security clearance. Dual U.S./Polish citizen.

**Languages:** English (native), Polish (fluent)

**Leadership:** Pi Kappa Phi — Pledge Class President (2025); Tech & Robotics Club President

**Aviation:** Student Pilot — FAA Private Pilot Knowledge Test (in progress); 6 hours in powered & 3 hours in glider flight training